

# Ankle Injuries Among United States High School Sports Athletes, 2005–2006

Alex J. Nelson, BA\*; Christy L. Collins, MA†; Ellen E. Yard, MPH†;  
Sarah K. Fields, JD, PhD\*; R. Dawn Comstock, PhD\*†

\*The Ohio State University, Columbus, OH; †Columbus Children's Research Institute, Children's Hospital, Columbus, OH

**Context:** Ankle injuries are the most common sport-related injuries. To date, no studies have been published that use national data to present a cross-sport, cross-sex analysis of ankle injuries among US high school athletes.

**Objective:** To investigate the incidence rates of ankle injuries by sex, type of exposure, and sport.

**Design:** Descriptive epidemiologic study.

**Setting:** One hundred US high schools.

**Patients or Other Participants:** United States high school athletes.

**Main Outcome Measure(s):** We reviewed ankle injury data collected over the 2005–2006 school year from a nationally representative sample obtained by High School RIO, an injury surveillance system. Specific sports studied were boys' football, boys' and girls' soccer, girls' volleyball, boys' and girls' basketball, boys' wrestling, boys' baseball, and girls' softball.

**Results:** An estimated 326 396 ankle injuries occurred nationally in 2005–2006, yielding an injury rate of 5.23 ankle injuries per 10 000 athlete-exposures. Ankle injuries occurred at

a significantly higher rate during competition (9.35 per 10 000 athlete-exposures) than during practice (3.63) (risk ratio = 2.58; 95% confidence interval = 2.26, 2.94;  $P < .001$ ). Boys' basketball had the highest rate of ankle injury (7.74 per 10 000 athlete-exposures), followed by girls' basketball (6.93) and boys' football (6.52). In all sports except girls' volleyball, rates of ankle injury were higher in competition than in practice. Overall, most ankle injuries were diagnosed as ligament sprains with incomplete tears (83.4%). Ankle injuries most commonly caused athletes to miss less than 7 days of activity (51.7%), followed by 7 to 21 days of activity loss (33.9%) and more than 22 days of activity loss (10.5%).

**Conclusions:** Sports that combine jumping in close proximity to other players and swift changes of direction while running are most often associated with ankle injuries. Future research on ankle injuries is needed to drive the development and implementation of more effective preventive interventions.

**Key Words:** injury surveillance, injury epidemiology, lower extremity injuries

## Key Points

- This is the first study to compare the epidemiology of ankle injuries across sports and between sexes among US high school athletes using nationally representative data.
- Among US high school athletes participating in 9 sports during the 2005–2006 school year, ankle injuries accounted for 22.6% of all injuries, with a total ankle injury rate of 5.23 injuries per 10 000 athlete-exposures.
- Of the high school sports studied, the 4 that involve jumping in close proximity to other players and swift changes of direction while running (football, basketball, soccer, and volleyball) were those with the highest rates of ankle injury.
- Studies are needed to enhance our understanding of ankle injuries in high school athletes and to drive the development and implementation of more effective targeted preventive interventions.

In the 2005–2006 school year, more than 7.1 million students in the United States (US) participated in high school athletics.<sup>1</sup> This represents an all-time high of 53.5% of enrolled students and demonstrates the 17th straight year of growth in high school athletic participation. Although athletics play an important role in an adolescent's adoption of a healthy lifestyle, including physical fitness, participation in any sport carries an inherent risk of injury. As high school athletic participation continues to grow, the incidence of sport-related injury is also expected to increase.

Patterns of sports-related injury vary by sport,<sup>2–4</sup> sex,<sup>2,3,5–7</sup> age,<sup>2,3,8,9</sup> and type of athletic exposure (eg, competition or practice).<sup>6,10</sup> Despite sport-specific patterns, one fact is consistent: of the nearly 2 000 000 injuries that occur yearly in

organized high school athletics,<sup>11</sup> the ankle is the most commonly reported body site of injury.<sup>1,2,4–8,10–17</sup> The high rate of ankle injuries among athletes has driven researchers to attempt to identify risk factors<sup>2–6,8,10–12,17,18</sup> and preventive methods.<sup>16,19–21</sup> Prior reports of ankle injuries among athletes have included single sports in a limited population,<sup>5,6,15,17</sup> evaluation of risk or preventive factors,<sup>16,19,20,21</sup> and general sport-related injury studies in which the ankle was only one of many injured sites.<sup>2–4,8–13,16,18,22</sup> None of these authors have presented a comparative epidemiology of ankle injuries across sports and between sexes among high school athletes using nationally representative data. Well-designed surveillance systems are needed to collect the accurate exposure and

injury data required to determine ankle injury rates across sports and between sexes.

Our goal was to describe the ankle injuries that occurred among high school athletes participating during the 2005–2006 academic year in 9 different sports using data drawn from a nationally representative sample of high schools.

## METHODS

Certified athletic trainers (ATs) who possessed current National Athletic Trainers' Association (NATA) membership and Board of Certification certification, provided care to high school athletes, and had valid e-mail addresses were invited to participate in the study as data reporters ( $n = 4120$ ). Schools with ATs willing to participate ( $n = 425$ ) were stratified into 8 sampling groups by 4 US Census geographic locations<sup>23</sup> and 2 school size categories ( $<1000$  versus  $\geq 1000$  students). A random sample of schools from each stratum (12 schools from 4 strata and 13 schools from the other 4 strata) was selected to obtain a representative study sample of 100 schools. To maintain the representativeness of the study sample, if a school dropped out of the study, another school from the same stratum was randomly selected for replacement.

The ATs at participating high schools reported data weekly throughout the 2005–2006 academic year using an Internet-based injury surveillance system, High School RIO (Reporting Information Online, Columbus Children's Research Institute, Columbus, OH), which collected athlete-exposures (A-Es) and injury and injury event information for all injuries sustained by high school athletes participating in 5 boys' sports (11-man football, soccer, basketball, wrestling, and baseball), and 4 girls' sports (soccer, volleyball, basketball, and fast-pitch softball). The RIO system, which has been used in several other studies of rugby and summer camp injuries,<sup>24</sup> has been described in detail previously.<sup>25</sup> The RIO system is closely modeled after the National Collegiate Athletic Association Injury Surveillance System, which has collected high-quality data on injuries, A-Es, and risk factors since 1982 and has been used to develop preventive interventions that have had proven success in reducing injuries among collegiate athletes.<sup>26</sup> A reportable injury was defined as one that occurred as a result of an organized high school athletic practice or competition, required medical attention by a team AT or a physician, and resulted in restriction of the athlete's participation for 1 or more days beyond the day of injury. An A-E was defined as 1 athlete participating in 1 practice or competition. Each week, reporters received an e-mail reminding them to log into the Internet-based surveillance system. Reporters who failed to log in to complete the weekly exposure and injury report received an e-mail or phone call reminding them to do so.

We used SPSS with the Complex Samples module (version 14.0; SPSS Inc, Chicago, IL) for data analysis. To calculate national estimates of the number of injuries, we assigned each reported injury a sample weight based on the inverse of the probability of the school's selection into the study (based on the total number of US high schools in each of the 8 sampling strata). Statistical analyses included the  $\chi^2$  test with Yates correction and  $t$  tests. We used rate ratios (RRs) and injury proportion ratios (PRs) as measures of the magnitude of associations. The following is an example of the RR calculation comparing the rate of competition ankle injuries to the rate of practice ankle injuries:

$$RR = (\text{number of competition ankle injuries/number of competition A-Es}) \times 10\,000$$

---


$$(\text{number of practice ankle injuries/number of practice A-Es}) \times 10\,000$$


---

The following is an example of how PRs were calculated:

$$PR = [(\text{national estimated No. girls' basketball ankle injuries while rebounding})/(\text{national estimated total No. girls' basketball ankle injuries})]$$

---


$$[(\text{national estimated No. boys' basketball ankle injuries while rebounding})/(\text{national estimated total No. boys' basketball ankle injuries})].$$

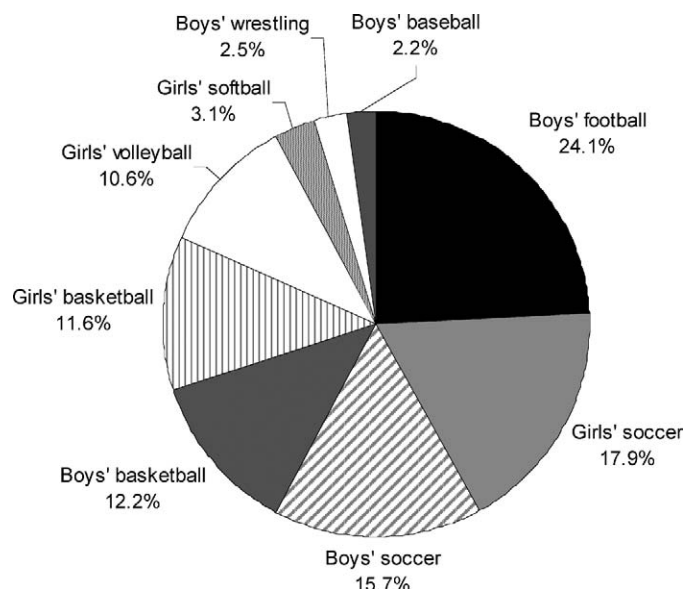

---

Statistical significance was assessed using 95% confidence intervals (CIs) and  $P$  values, with  $P < .05$  considered significant. All data presented reflect national estimates, unless otherwise stated. This study was approved by the Columbus Children's Research Institute's Institutional Review Board. We were granted a waiver of the informed consent/assent requirement under the Institutional Review Board Latitude to Approve a Consent Procedure that Alters or Waives Some or All of the Elements of Consent, §46.116.

## RESULTS

In the 2005–2006 school year, athletes from the 9 sports of interest sustained 905 ankle injuries during 1 730 764 A-Es, for an ankle injury rate of 5.23 ankle injuries per 10 000 A-Es. This number represents an estimated 326 396 ankle injuries sustained nationally (22.6% of all high school sports-related injuries). Ankle injury rates by sport, type of exposure (competition versus practice), and sex are presented in the Table. Overall, ankle injury rates were higher in competitions than in practices ( $RR = 2.58$ ; 95% CI = 2.26, 2.94;  $P < .001$ ). This finding was consistent in all sports except volleyball (Table). Overall, girls had an ankle injury rate that was similar to boys (5.39 versus 5.15 per 10 000 A-Es). In sports played by both sexes (soccer, basketball, and baseball or softball), boys had higher rates of practice-related ankle injuries, but girls had higher rates of competition-related ankle injuries. Despite having a lower ankle injury rate than both boys' and girls' basketball, football accounted for the greatest portion of all high school ankle injuries (24.1%), followed by boys' and girls' soccer (Figure 1). Wrestling, baseball, and softball will not be discussed further as each had an ankle injury rate of less than 2.0 injuries per 10 000 A-Es, and combined, they accounted for less than 10% of all ankle injuries.

As shown in Figure 2, injuries to the ankle constitute a higher proportion of all injuries sustained by female athletes (32.5%) than by male athletes (18.4%) ( $PR = 1.77$ ; 95% CI = 1.53, 2.05;  $P < .001$ ). Ankle injuries occurred slightly more often among seniors (29.7%) than among juniors (25.2%), sophomores (26.0%), or freshmen (19.1%). No significant differences were noted in proportions of ankle injuries by height, weight, or body mass index of injured high school athletes. Ankle injuries were most frequently diagnosed as ligament sprains with incomplete tears (83.4%), followed by diagnoses of fractures (5.2%), ligament sprains with complete tears (4.0%), and contusions (2.0%) (Figure 3). A total of 81.8% of ankle injuries were new, 9.4% were recurrent injuries incurred



**Figure 1.** Ankle injuries by high school sport, High School Sports-Related Injury Surveillance Study, United States, 2005–2006 school year.

during the same season as the initial injury, 8.1% were recurrences of injuries initially incurred during a prior season, and 0.7% were unknown or other. The athlete was wearing an ankle brace when 7.8% of ankle injuries occurred.

The outcome of ankle injuries, measured as the length of time an athlete was unable to participate in practice or competition (or both), is presented in Figure 4. Ankle injuries most commonly caused athletes to miss less than 7 days of activity (51.7%), followed by 7 to 21 days of activity (33.9%), more than 22 days of activity (10.5%), and “other/unknown” (3.9%). By multiplying the median of the days lost in each category by the number of athletes in each category, we calculated that ankle injuries were responsible for an estimated 2 287 536 days of activity lost. Of those ankle injuries that occurred during competition ( $n = 172\,118$ ), 5.3% were directly related to action that was ruled as illegal or as foul play by a referee or disciplinary committee.

## Football

Football accounted for 24.1% of all ankle injuries (Figure 1). The most common mechanism of football ankle injury was contact with another person (61.2% of football ankle injuries), followed by contact with the playing surface (28.2%). More specifically, ankle injuries most often occurred while tackling (45.8%), blocking (18.6%), and being stepped on (12.6%). The types of plays most frequently associated with ankle injuries were running plays (61.9%), followed by passing plays (15.5%), special teams plays (11.9%), and general play or conditioning (6.4%); 4.3% were other/unknown. Ankle injuries were most commonly sustained by running backs (26.4% of all football ankle injuries), followed by wide receivers and flankers (11.6%). Less than 1% of football ankle injuries sustained in competition (0.7%) resulted from illegal or foul play. A total of 81.8% of football ankle injuries were new injuries. The athlete was wearing an ankle brace in 10.8% of football ankle injuries.

## Soccer

Soccer accounted for 33.6% of all ankle injuries (boys = 15.7%, girls = 17.9%) (Figure 1). Injuries to the ankle were the most frequent injury in both boys' and girls' soccer, but the proportion of ankle injuries to total soccer injuries was higher among girls (31.5%) than it was among boys (23.5%;  $PR = 1.35$ ; 95%  $CI = 1.00, 1.81$ ;  $P < .05$ ) (Figure 2). Fractures made up a higher proportion of soccer injuries among girls (11.3%) than among boys (2.0%), although the difference was not statistically significant ( $PR = 5.55$ ; 95%  $CI = 0.70, 43.93$ ;  $P = .06$ ). A total of 77.3% of boys' soccer ankle injuries and 85.2% of girls' soccer ankle injuries were new injuries. None of the boys who sustained ankle injuries were wearing ankle braces, whereas 2.5% of the girls who sustained ankle injuries were wearing ankle braces.

Injury mechanisms related to a higher proportion of ankle injuries in boys' soccer than in girls' soccer included contact with another person (47.1% versus 40.8%, respectively;  $PR = 1.16$ ; 95%  $CI = 0.78, 1.72$ ;  $P = .48$ ) and contact with the playing surface (41.1% versus 31.4%, respectively;  $PR = 1.31$ ; 95%  $CI = 0.82, 2.09$ ;  $P = .28$ ). Conversely, a higher proportion of ankle injuries in girls' soccer than in boys' soccer was due to contact with the playing apparatus (eg, ball, goalpost) (11.7% versus 4.1%, respectively;  $PR = 2.84$ ; 95%  $CI = 0.70, 11.62$ ;  $P = .12$ ) and no contact (ie, rotation around a planted foot, etc) (11.3% versus 4.1%, respectively;  $PR = 2.30$ ; 95%  $CI = 0.89, 8.50$ ;  $P = .07$ ), although these sex differences were not statistically significant. A higher proportion of ankle injuries in boys' soccer than in girls' soccer were due to general play (22.4% versus 12.6%, respectively;  $PR = 1.78$ ; 95%  $CI = 0.85, 3.73$ ;  $P = .12$ ) and heading the ball (4.4% versus 1.1%, respectively;  $PR = 3.98$ ; 95%  $CI = 0.41, 38.86$ ;  $P = .20$ ), although these sex differences were not statistically significant. A greater proportion of boys' soccer ankle injuries sustained in competition were ruled illegal or foul play than in girls' soccer (8.5% versus 6.0%, respectively;  $PR = 1.41$ ; 95%  $CI = 0.63, 3.15$ ;  $P = .32$ ), although, again, the difference was not statistically significant.

## Volleyball

Girls' volleyball accounted for 10.6% of all high school ankle injuries (Figure 1). Of the sports described in this study, volleyball had the highest proportion of ankle injuries (42.4%; Figure 2). The most common mechanisms associated with ankle injuries in volleyball were contact with another person (48.7% of total volleyball ankle injuries) and contact with the playing surface (41.6%). More specifically, ankle injuries most often occurred during blocking (51.0%), spiking (14.2%), passing (13.0%), and general play (6%). Ankle injuries most frequently occurred to the middle blocker (40.2% of all ankle injuries). The court locations at which the highest proportions of ankle injuries took place were middle forward (32.5%), right forward (20.7%), and left forward (14.4%). A total of 91.4% of volleyball ankle injuries were new injuries. In 10.4% of volleyball ankle injuries, the athlete was wearing an ankle brace.

## Basketball

Basketball accounted for 23.8% of all ankle injuries (boys = 12.2%, girls = 11.6%; Figure 1). Boys' basketball had the highest ankle injury rate per 10 000 A-Es of all sports studied



**Ankle Injury Rates per 10 000 Athlete-Exposures by Sport, Sex, and Type of Exposure: High School Sport-Related Injury Surveillance Study, United States, 2005–2006 School Year**

| Sport                | Exposure Type | Ankle Injuries,<br>No. | Athlete-Exposures | Injury Rates<br>per 10 000<br>Athlete-Exposures |
|----------------------|---------------|------------------------|-------------------|-------------------------------------------------|
| <b>Boys' sports</b>  |               |                        |                   |                                                 |
| Football             | Practice      | 130                    | 349 183           | 3.72                                            |
|                      | Competition   | 151                    | 82 059            | 18.40                                           |
|                      | <b>Totals</b> | <b>281</b>             | <b>431 242</b>    | <b>6.52</b>                                     |
| Soccer               | Practice      | 40                     | 104 106           | 3.84                                            |
|                      | Competition   | 43                     | 49 294            | 8.72                                            |
|                      | <b>Totals</b> | <b>83</b>              | <b>153 400</b>    | <b>5.41</b>                                     |
| Basketball           | Practice      | 101                    | 156 679           | 6.45                                            |
|                      | Competition   | 68                     | 61 663            | 11.03                                           |
|                      | <b>Totals</b> | <b>169</b>             | <b>218 342</b>    | <b>7.74</b>                                     |
| Wrestling            | Practice      | 19                     | 126 059           | 1.51                                            |
|                      | Competition   | 12                     | 40 220            | 2.98                                            |
|                      | <b>Totals</b> | <b>31</b>              | <b>166 279</b>    | <b>1.86</b>                                     |
| Baseball             | Practice      | 11                     | 115 564           | 0.95                                            |
|                      | Competition   | 16                     | 63 871            | 2.51                                            |
|                      | <b>Totals</b> | <b>27</b>              | <b>179 435</b>    | <b>1.50</b>                                     |
| <b>Girls' sports</b> |               |                        |                   |                                                 |
| Soccer               | Practice      | 32                     | 98 166            | 3.26                                            |
|                      | Competition   | 58                     | 43 415            | 13.36                                           |
|                      | <b>Totals</b> | <b>90</b>              | <b>141 581</b>    | <b>6.36</b>                                     |
| Volleyball           | Practice      | 49                     | 75 544            | 6.49                                            |
|                      | Competition   | 25                     | 43 691            | 5.72                                            |
|                      | <b>Totals</b> | <b>74</b>              | <b>119 235</b>    | <b>6.21</b>                                     |
| Basketball           | Practice      | 62                     | 132 836           | 4.67                                            |
|                      | Competition   | 67                     | 53 325            | 12.56                                           |
|                      | <b>Totals</b> | <b>129</b>             | <b>186 161</b>    | <b>6.93</b>                                     |
| Softball             | Practice      | 8                      | 88 362            | 0.91                                            |
|                      | Competition   | 13                     | 46 727            | 2.78                                            |
|                      | <b>Totals</b> | <b>21</b>              | <b>135 089</b>    | <b>1.55</b>                                     |
| Boys' total          | Practice      | 301                    | 851 591           | 3.53                                            |
|                      | Competition   | 290                    | 297 107           | 9.76                                            |
|                      | <b>Totals</b> | <b>591</b>             | <b>1 148 698</b>  | <b>5.14</b>                                     |
| Girls' total         | Practice      | 151                    | 394 908           | 3.82                                            |
|                      | Competition   | 163                    | 187 158           | 8.71                                            |
|                      | <b>Totals</b> | <b>314</b>             | <b>582 066</b>    | <b>5.39</b>                                     |
| Combined totals      | Practice      | 452                    | 1 246 499         | 3.63                                            |
|                      | Competition   | 453                    | 484 265           | 9.35                                            |
|                      | <b>Totals</b> | <b>905</b>             | <b>1 730 764</b>  | <b>5.23</b>                                     |

(7.74), and although the sex difference was not statistically significant, boys' basketball players had a higher ankle injury rate than did girls' basketball players (6.93). Ankle injuries represented a slightly higher proportion of all injuries in boys' basketball than in girls' basketball (39.7% versus 36.5%, respectively).

A greater proportion of ankle injuries in boys' basketball tended to be related to contact with another person than in girls' basketball, although this difference was not significant (60.1% versus 53.6%, respectively;  $PR = 1.12$ ; 95%  $CI = 0.88, 1.43$ ;  $P = .38$ ). Although rebounding was the activity associated with the greatest proportion of ankle injuries in both boys' and girls' basketball (47.1% and 31.9%, respectively), the sex difference was significant ( $PR = 1.47$ ; 95%  $CI = 1.04, 2.10$ ;  $P = .03$ ). Additional activities associated with ankle injuries in boys' and girls' basketball were shooting the ball (12.1% versus 8.0%, respectively), ball handling/dribbling (7.3% versus 14.0%, respectively), defending (12.2% versus 12.8%, respectively), and general play (10.5% versus 18.1%, respectively). In boys' basketball, a greater proportion of all

ankle injuries was sustained by forwards (the primary rebounding position) than in girls' basketball (46.2% versus 26.0%, respectively;  $PR = 1.78$ ; 95%  $CI = 1.21, 2.62$ ;  $P < .01$ ). Conversely, in girls' basketball, a greater proportion of ankle injuries tended to occur to guards than in boys' basketball (49.2% versus 41.8%, respectively;  $PR = 1.18$ ; 95%  $CI = 0.88, 1.58$ ;  $P = .29$ ). Although the sex differences were not statistically significant, a greater proportion of ankle injuries tended to occur in the inside lane (where most rebounding occurs) in boys' basketball than in girls' basketball (67.7% versus 56.5%, respectively;  $PR = 1.20$ ; 95%  $CI = 0.96, 1.50$ ;  $P = .10$ ), and a greater proportion of ankle injuries tended to occur between the 3-point arc and the lane (where guards play) in girls' basketball than in boys' basketball (22.5% versus 15.0%, respectively;  $PR = 1.50$ ; 95%  $CI = 0.84, 2.68$ ;  $P = .17$ ). Although no girls' basketball injuries were related to illegal or foul play, 2.5% of boys' basketball ankle injuries sustained in competition were related to illegal or foul play. A total of 78.1% of boys' and 74.3% of girls' basketball injuries were new injuries. In 12.6% of the boys' injuries, the athlete

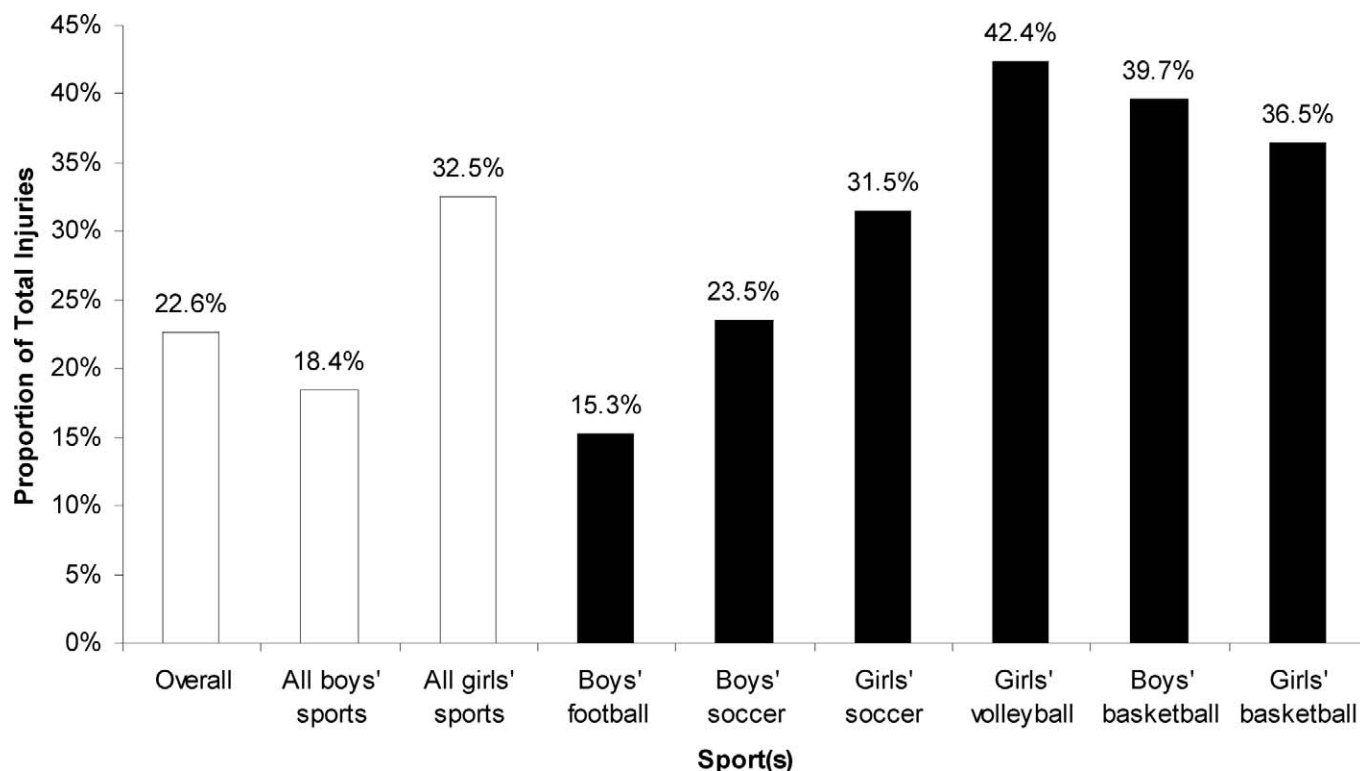


Figure 2. Ankle injuries as a proportion of total high school injuries by sport, High School Sports-Related Injury Surveillance Study, United States, 2005–2006 school year.

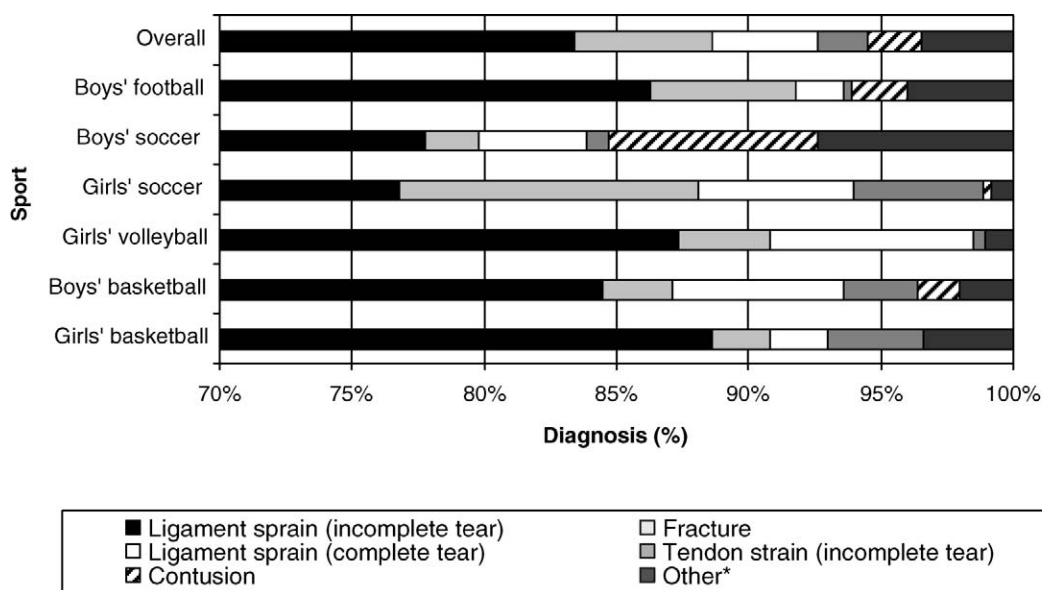


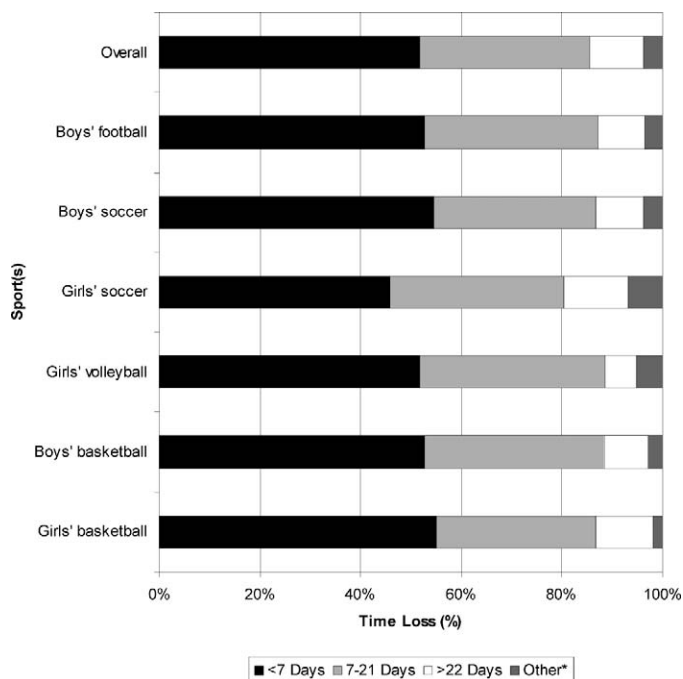
Figure 3. Diagnosis of ankle injuries by high school sport, High School Sports-Related Injury Surveillance Study, United States, 2005–2006 school year. \*\*Other\*\* includes diagnoses such as stress fracture, muscle strain, tendinitis, etc.

was wearing an ankle brace, compared with 18.6% of the girls' injuries.

## DISCUSSION

Ankle injuries are the most common injuries sustained by high school athletes. As participation in high school athletics continues to increase, so will the number of athletes who sustain ankle injuries. In this surveillance study of injuries among

US high school athletes participating in boys' football, girls' and boys' soccer, girls' volleyball, girls' and boys' basketball, boys' wrestling, boys' baseball, and girls' softball during the 2005–2006 school year, ankle injuries accounted for 22.6% of all injuries, with a total ankle injury rate of 5.23 injuries per 10 000 A-Es. Incidence rates of ankle injuries varied by sport, sex, and type of athletic exposure (competition versus practice). This study is the first to present ankle injury data from a nationally representative sample of US high schools by sport,



**Figure 4. Time loss for ankle injuries by high school sport, High School Sports-Related Injury Surveillance Study, United States, 2005–2006 school year. \*Other includes medical disqualification for the season, situations in which the athlete chose not to continue, etc.**

sex, and exposure type. Because we used the same definition of injury, units of exposure, and reporting system, we can directly compare ankle injuries across sports and between sexes, a task that has been difficult in the past because of variability of injury definitions and units of exposure across studies.

Ankle injuries were more common in the sports that involved both running and jumping, with the highest rates occurring in football, basketball, soccer, and volleyball. This finding is consistent with the results of prior studies.<sup>5,6,15,17,19</sup> Activities that involve jumping in close proximity to other players were also associated with ankle injuries (ie, rebounding and shooting in basketball and spiking and blocking in volleyball). Other types of activity that were related to ankle injury involved swift changes of direction while running: for example, the activities required of the running back and wide receiver positions in football and the general play of all positions in soccer and basketball.

Prior authors have reported varied results regarding sex differences in ankle injuries. Some have noted higher rates among male athletes,<sup>6,10</sup> whereas others have reported higher rates among female athletes.<sup>2,7</sup> We found a similar ankle injury rate per 10 000 A-Es among high school girls and boys. Ankle injuries represented a greater proportion of all body sites injured among girls than among boys (31.7% versus 18.3%). However, this sex difference varied by type of athletic exposure. Girls had higher rates of practice-related ankle injuries than did boys (3.82 versus 3.53 per 10 000 A-Es), but boys had higher rates of competition-related injuries (9.76 versus 8.71 per 10 000 A-Es).

In all sports except girls' volleyball, rates of ankle injury were greater during competition than during practice, which is consistent with the results of previous studies.<sup>6,10</sup> The inherent increase in intensity in competition situations is one logical

explanation. Furthermore, our data demonstrate that certain locations on the playing field/court were associated with higher rates of ankle injuries. For example, a greater percentage of ankle injuries occurred in the lane in basketball and at the net in volleyball.

The fact that jumping in close proximity to other players and changing direction while running are major components of both boys' and girls' basketball likely explains our findings that these sports had the highest ankle injury rates. These results are consistent with those of previous authors, who noted that rebounding was the activity most often associated with ankle injuries in high school basketball players<sup>10</sup> and that basketball players were most likely to sustain ankle injuries.<sup>10,16,21</sup>

Past investigators<sup>8,9</sup> have suggested that an adolescent is more likely than a prepubescent to sustain an ankle injury, given the development of greater body mass and strength during this period of maturation. The body mass and strength of many student-athletes continue to increase throughout high school, which may account for our findings that the fewest ankle injuries occurred among freshmen and the greatest number occurred among seniors. However, based on our data, ankle injuries did not seem to vary significantly by height, weight, or body mass index. An alternative explanation may be that the older students were involved in more intense competitions (ie, varsity versus junior varsity or freshman games). Additional studies are needed to further examine differences in ankle injuries based on age, height, weight, and body mass.

As do all studies, this study has its limitations. Given that eligible schools had to have an AT who was a member of the NATA on staff, our results may not be generalizable to all schools in the US. The immediate attention to injury that an AT provides could aid in diagnosis, treatment, and prevention of further injury, thus reducing injury rates. Alternatively, the presence of the AT may result in an increase in reported injury rates from enhanced diagnosis and reporting. However, we felt this potential limitation was outweighed by the increased quality of data provided by these medically trained professionals. Additional socioeconomic differences may exist between schools with and without the services of ATs, which could result in differences involving other potential risk factors, such as field condition or quality of equipment.

Despite the current movement to use a more quantitative, time-based unit of A-E (such as minutes or hours),<sup>27</sup> our definition of 1 athlete participating in 1 practice or competition was the most feasible definition of A-E for our study, given that our data reporters, high school ATs, could not necessarily be present to time the participation of every athlete in every sport. Additionally, we contend that every athlete is at risk for injury during every practice and competition in which he or she participates. Other limitations of this study relate to our inability to track individual athletes across several seasons to capture recurrent injuries, as well as the limited usefulness of the information we collected on protective equipment, when past authors<sup>19,20</sup> have shown that ankle braces and taping may protect athletes from ankle-related injuries.

It is often assumed that ankle injuries are an unavoidable part of sport participation because of their high frequency. The prevalent morbidity and burden on health care that results from ankle injuries have been discussed in many publications.<sup>7,20,21,28–32</sup> Such negative outcomes could be reduced by identifying those athletes most at risk for ankle injuries and by implementing targeted preventive measures. Of the high school sports studied, the 4 that involve activities of jumping

in close proximity to other players and swift changes of direction while running (football, basketball, soccer, and volleyball) were those most often associated with injuries to the ankle. To enhance our understanding of ankle injuries in high school athletes, future researchers could combine the reporting power of High School RIO with the preseason studies and tracking that have been used in prior studies, such as preseason evaluation of postural sway,<sup>21</sup> ankle strength and flexibility,<sup>21</sup> preseason compilation balance scores,<sup>16</sup> and presence of particular types of ankle braces and their resultant effectiveness.<sup>19,20</sup> Future research will add to our growing knowledge of the patterns of ankle injuries and, thus, will drive the development and implementation of more effective preventive interventions.

## ACKNOWLEDGMENTS

The lead author was funded by the Samuel J. Roessler Scholarship from the Landacre Honor Society at The Ohio State University College of Medicine. This study was funded by the Centers for Disease Control and Prevention (CDC), grant R49/CE000674. The content of this report is solely the responsibility of the authors and does not necessarily represent the official views of the CDC.

## REFERENCES

- National Federation of State High School Associations. Participation in high school sports increases again; confirms NFHS commitment to stronger leadership. October 2006. Available at: [http://www.nfhs.org/web/2006/09/participation\\_in\\_high\\_school\\_sports\\_increases\\_again\\_confirms\\_nf.aspx](http://www.nfhs.org/web/2006/09/participation_in_high_school_sports_increases_again_confirms_nf.aspx). Accessed December 1, 2006.
- Beynon BD, Vacek PM, Murphy D, Alosa D, Paller D. First-time inversion ankle ligament trauma: the effects of sex, level of competition, and sport on the incidence of injury. *Am J Sports Med*. 2005;33:1485–1491.
- DeHaven KE, Lintner DM. Athletic injuries: comparison by age, sport, and gender. *Am J Sports Med*. 1986;14:218–224.
- McLain LG, Reynolds S. Sports injuries in a high school. *Pediatrics*. 1989;84:446–450.
- McKay GD, Goldie PA, Payne WR, Oakes BW, Watson LF. A prospective study of injuries in basketball: a total profile and comparison by gender and standard of competition. *J Sci Med Sport*. 2001;4:196–211.
- Messina DF, Farney WC, DeLee JC. The incidence of injury in Texas high school basketball: a prospective study among male and female athletes. *Am J Sports Med*. 1999;27:294–299.
- Richards CF. Ankle injury, soft tissue. January 23, 2006. Available at: <http://www.emedicine.com/EMERG/topic30.htm>. Accessed November 28, 2006.
- Adirim TA, Cheng TL. Overview of injuries in the young athlete. *Sports Med*. 2003;33:75–81.
- Taylor BL, Attia MW. Sports-related injuries in children. *Acad Emerg Med*. 2000;7:1376–1382.
- Powell JW, Barber-Foss KD. Sex-related injury patterns among selected high school sports. *Am J Sports Med*. 2000;28:385–391.
- Powell JW, Barber-Foss KD. Injury patterns in selected high school sports: a review of the 1995–1997 seasons. *J Athl Train*. 1999;34:277–284.
- Pontell D, Hallivis R, Dollard MD. Sports injuries in the pediatric and adolescent foot and ankle: common overuse and acute presentations. *Clin Podiatr Med Surg*. 2006;23:209–231.
- Centers for Disease Control and Prevention (CDC). Nonfatal sports- and recreation-related injuries treated in emergency departments: United States, July 2000–June 2001. *MMWR Morb Mortal Wkly Rep*. 2002;51:736–740.
- Anandacoomarasamy A, Barnsley L. Long term outcomes of inversion ankle injuries. *Br J Sports Med*. 2005;39:e14.
- Greene JJ, McGuine TA, Levenson G, Best TM. Anthropometric and performance measures for high school basketball players. *J Athl Train*. 1998;33:229–232.
- McGuine TA, Greene JJ, Best T, Levenson G. Balance as a predictor of ankle injuries in high school basketball players. *Clin J Sport Med*. 2000;10:239–244.
- Verhagen EA, Van der Beek AJ, Bouter LM, Bahr RM, Van Mechelen W. A one season prospective cohort study of volleyball injuries. *Br J Sports Med*. 2004;38:477–481.
- Cheng TL, Fields CB, Brenner RA, Wright JL, Lomax T, Scheidt PC. Sports injuries: an important cause of morbidity in urban youth. District of Columbia Child/Adolescent Injury Research Network. *Pediatrics*. 2000;105:E32.
- Kinzy SJ, Ingersoll CD, Knight KL. The effects of selected ankle appliances on postural control. *J Athl Train*. 1997;32:300–303.
- Pienkowski D, McMorrow M, Shapiro R, Caborn DN, Stayton J. The effect of ankle stabilizers on athletic performance: a randomized prospective study. *Am J Sports Med*. 1995;23:757–762.
- Wang HK, Chen CH, Shiang TY, Jan MH, Lin KH. Risk-factor analysis of high school basketball-player ankle injuries: a prospective controlled cohort study evaluating postural sway, ankle strength, and flexibility. *Arch Phys Med Rehabil*. 2006;87:821–825.
- Burt CW, Overpeck MD. Emergency visits for sports-related injuries. *Ann Emerg Med*. 2001;37:301–308.
- Regions of the United States. United States Census Bureau. Available at: <http://www.census.gov/const/regionmap.pdf>. Accessed February 13, 2007.
- Yard EE, Scanlin MM, Ebner Erceg L, et al. Illness and injury among children attending summer camp in the United States, 2005. *Pediatrics*. 118:e1342–e1349.
- Centers for Disease Control and Prevention. Sports-related injuries among high school athletes: United States, 2005–06 school year. *MMWR Morbid Mortal Wkly Rep*. 2006;55:1037–1040.
- NCAA. Injury Surveillance System. National Collegiate Athletic Association. Available at: <http://www.ncaa.org/iss.html>. Accessed February 9, 2007.
- Fuller CW, Ekstrand J, Junge A, et al. Consensus statement on injury definitions and data collection procedures in studies of football (soccer) injuries. *Clin J Sport Med*. 2006;16:97–106.
- Norkus SA, Floyd RT. The anatomy and mechanisms of syndesmotic ankle sprains. *J Athl Train*. 2001;36:68–73.
- American Academy of Orthopaedic Surgeons. Foot and ankle. July 2001. Available at: [http://orthoinfo.aaos.org/fact/thr\\_report.cfm?ThreadID=100&topcategory=Foot](http://orthoinfo.aaos.org/fact/thr_report.cfm?ThreadID=100&topcategory=Foot). Accessed December 14, 2006.
- Bridgman SA, Clement D, Downing A, Walley G, Phair I, Maffulli N. Population based epidemiology of ankle sprains attending accident and emergency units in the West Midlands of England, and a survey of UK practice for severe ankle sprains. *Emerg Med J*. 2003;20:508–510.
- Gehrig L, Miller B, Fehr N. Sprained ankle. March 2005. Available at: [http://orthoinfo.aaos.org/fact/thr\\_report.cfm?thread\\_id=152&topcategory=Foot](http://orthoinfo.aaos.org/fact/thr_report.cfm?thread_id=152&topcategory=Foot). Accessed December 11, 2006.
- Trevino SG, Davis P, Hecht PJ. Management of acute and chronic lateral ligament injuries of the ankle. *Orthop Clin North Am*. 1994;25:1–16.

Alex J. Nelson, BA, contributed to analysis and interpretation of the data and drafting, critical revision, and final approval of the article. Christy L. Collins, MA, contributed to conception and design; acquisition and analysis and interpretation of the data; and drafting, critical revision, and final approval of the article. Ellen E. Yard, MPH, contributed to conception and design, acquisition of the data, and critical revision and final approval of the article. Sarah K. Fields, JD, PhD, contributed to conception and design and drafting, critical revision, and final approval of the article. R. Dawn Comstock, PhD, contributed to conception and design; acquisition and analysis and interpretation of the data; and drafting, critical revision, and final approval of the article.

Address correspondence to Christy L. Collins, MA, Center for Injury Research and Policy, Columbus Children's Hospital, 700 Children's Drive, Columbus, OH 43205. Address e-mail to [collinsc@ccri.net](mailto:collinsc@ccri.net).